

Class 7 Computer Science (Click and Learn - 7) — SA1 Master Solutions & Complete Guide

Summative Assessment - 1 Complete Curriculum Blueprint (Chapters 1 to 4 & Test Paper - 1)

Syllabus Scope (SA-1 Timetable & Portions Covered):

1. **Chapter 1: Formulae and Functions in Excel 2016** (Pages 1–15)
2. **Chapter 2: Charts in Excel 2016** (Pages 16–29)
3. **Chapter 3: Advanced Features of Excel 2016** (Pages 30–41)
4. **Chapter 4: Introduction to Adobe Animate 2020** (Pages 42–55)
5. **Test Paper - 1: Comprehensive Assessment on Chapters 1 to 4** (Pages 56–58)

Pedagogical Standards:

- Every exercise, multiple-choice question (MCQ), fill-in-the-blank, true/false statement, matching column, practical step, formula breakdown, and differentiated table is 100% solved with zero omissions.
- Detailed step-by-step explanations, syntax breakdowns, ⚠️ Ninja Exam Traps, and ✅ Sanity Checks included for every single technical concept.

Chapter 1: Formulae and Functions in Excel 2016 (Pages 1–15)

Core Theoretical Concepts:

1. **What is a Formula?**
2. A formula is an expression that calculates values in a cell or range of cells.
3. **Golden Rule:** Every formula in Excel **MUST begin with an equal sign (=)**. Without =, Excel treats the entry as plain text.
4. **Elements of a Formula:**
 - Mathematical Operators: + (Addition), - (Subtraction), * (Multiplication), / (Division), ^ (Exponentiation), % (Percentage).
 - Cell References (Addresses): e.g., A1, B5, C10.
 - Constants: Numbers or fixed text values directly entered (e.g., =A1 * 10).
 - Functions: Pre-built formula commands (e.g., SUM, AVERAGE).
5. **Cell Ranges:**
6. A cell range is a group or block of contiguous cells selected in a worksheet.
7. Range notation uses a **colon (:)** between the top-left cell and bottom-right cell.
8. Types of Cell Ranges:
 - **Vertical Range:** Cells located in the same column (e.g., A1:A10).

- **Horizontal Range:** Cells located in the same row (e.g., A1:G1).
 - **Rectangular Range:** A 2D block across multiple rows and columns (e.g., B2:D8).
9. **Cell Referencing Modes:**
 10. **Relative Referencing:** Default mode. Cell references adjust automatically when copied to other cells (e.g., =A1+B1 copied down becomes =A2+B2).
 11. **Absolute Referencing:** Cell references stay locked to an exact location when copied. Identified by **dollar signs (\$)** before both column and row (e.g., =\$A\$1).
 12. **Mixed Referencing:** Either the column is locked or the row is locked (e.g., =\$A1 locks column A; =A\$1 locks row 1).
 13. **Functions & Basic Syntax:**
 14. **Syntax:** =Function_Name(Argument1, Argument2, ...)
 15. **Common Functions:**
 - =SUM(A1:A5): Adds all numbers in the specified range.
 - =AVERAGE(A1:A5): Calculates the arithmetic mean.
 - =COUNT(A1:A10): Counts cells containing numbers.
 - =MAX(A1:A10): Finds the highest numerical value in the range.
 - =MIN(A1:A10): Finds the lowest numerical value in the range.
 - =PRODUCT(A1:A3): Multiplies all numbers in the specified range.
 - =TODAY(): Returns the current system date.
 16. **AutoSum Feature:** Located on both the **Home Tab** (Editing Group) and **Formulas Tab**. Automatically detects adjacent cell ranges and inserts =SUM(). Shortcut: A1t + =.
 17. **Common Excel Error Codes:**
 18. #####: Column width is too narrow to display the number, or negative date/time. (Fix: Increase column width).
 19. #DIV/0!: Formula attempts to divide a number by zero or an empty cell.
 20. #VALUE!: Formula contains incorrect argument data type (e.g., adding text to a number: =A1 + "Apple").
 21. #NAME?: Function name is misspelled (e.g., =SUMM(A1:A5) instead of =SUM).
 22. #REF!: Invalid cell reference (occurs when referenced cells were deleted).
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Textbook Exercises — Chapter 1

A. Tick (✓) the correct option:

1. **Which symbol must every Excel formula begin with?**
2. a. + **b. = [✓]** c. @ d. #
3. **Which function is used to add numbers in a range?**
4. a. MAX b. AVERAGE **c. SUM [✓]** d. COUNT
5. **What is the symbol used for multiplication in Excel?**
6. a. x b. . **c. * [✓]** d. &
7. **Which function returns the largest value in a selected cell range?**
8. **a. MAX [✓]** b. MIN c. SUM d. LARGE
9. **The cell range B2:E6 contains how many total cells?**

10. a. 15 **b. 20 [✓]** c. 12 d. 24

(Working: 4 columns [B, C, D, E] $\times$ 5 rows [2, 3, 4, 5, 6] = 20 cells).

B. Fill in the blanks:

1. A formula in Excel always begins with an **equal to (=)** sign.
2. The colon (:) symbol is used to separate the starting and ending cell addresses of a range.
3. The **AutoSum** feature allows quick addition of numbers without typing the full formula manually.
4. The **PRODUCT** function multiplies all numbers given as arguments.
5. In **absolute** referencing, a dollar sign (\$) is placed before both column letter and row number.

C. State True (T) or False (F):

1. =A1+B1 is an example of an absolute reference. $\rightarrow$ **[F]** (It is a relative reference).
2. Functions in Excel are pre-designed formulas to perform complex calculations. $\rightarrow$ **[T]**
3. #DIV/0! error appears when a column is not wide enough. $\rightarrow$ **[F]** (It appears when divided by zero; ##### appears when column is narrow).
4. The COUNT function counts both text and numeric cells. $\rightarrow$ **[F]** (It counts only cells containing numerical values).
5. The TODAY() function requires no arguments inside its parentheses. $\rightarrow$ **[T]**

D. Answer the following questions in brief:

Q1: Define a Cell Range with an example.

Answer: A cell range is a collection or block of two or more adjacent cells selected in an Excel worksheet. It is represented by the starting cell address and ending cell address separated by a colon (:).

Example: B2:D6 refers to the rectangular block of cells spanning columns B to D and rows 2 to 6.

Q2: What is the basic syntax of an Excel function?

Answer: The basic syntax is:

=FUNCTION_NAME(Argument1, Argument2, ...)

Example: =SUM(A1:A10) where =SUM is the function name and A1:A10 is the argument range.

Q3: Explain the difference between MAX() and MIN() functions.

Answer:

- MAX() returns the highest (maximum) numerical value from the specified range (e.g., =MAX(10, 45, 22) returns 45).

- MIN() returns the lowest (minimum) numerical value from the specified range (e.g., =MIN(10, 45, 22) returns 10).

Q4: What causes the #NAME? error in Excel?

Answer: The #NAME? error occurs when Excel does not recognize text entered in a formula, most commonly caused by misspelling a function name (e.g., typing =SUUM(A1:A5)) or referencing an undefined range name.

E. Answer the following questions in detail:

Q1: Explain the three types of cell referencing in Excel 2016 with suitable examples.

Answer:

1. **Relative Referencing:** The cell address in the formula changes automatically when copied to another cell based on relative position.

Example: If cell C1 contains =A1+B1 and is copied down to C2, it automatically becomes =A2+B2.

2. **Absolute Referencing:** The cell address remains locked and does not change when copied. It uses dollar signs before both column and row (\$A\$1).

Example: In =B2 * \$C\$1, when copied to row 3, it becomes =B3 * \$C\$1. Cell C1 remains constant.

3. **Mixed Referencing:** Either the column is locked or the row is locked, but not both.

Example: =\$A1 (locks column A, row changes) or =A\$1 (locks row 1, column changes).

Chapter 2: Charts in Excel 2016 (Pages 16–29)

Core Theoretical Concepts:

1. **What is a Chart?**
2. A chart is a graphical, pictorial representation of numeric worksheet data.
3. Benefits: Visualizes trends, simplifies complex data, enables instant comparison, and makes presentations impactful.
4. **Components of a Chart:**
5. **Chart Area:** The entire chart window including all objects, titles, legends, and plot area.
6. **Plot Area:** The bounded rectangular area where the actual data is plotted, surrounded by axes.
7. **Chart Title:** Descriptive heading placed at the top explaining what the chart represents.
8. **Category Axis (X-Axis):** The horizontal axis displaying data categories (e.g., Student Names, Months).
9. **Value Axis (Y-Axis):** The vertical axis displaying numerical values/measurements (e.g., Marks, Sales).
10. **Data Series:** The actual visual bars, slices, columns, or lines that represent individual data points.
11. **Legend:** A color-coded or patterned key identifying which color/symbol corresponds to which data series.
12. **Gridlines:** Horizontal and vertical lines extending across the plot area to make reading values easier.

13. **Data Labels:** Labels that show the exact numerical values directly on top of the bars or data markers.
 14. **Types of Charts in Excel 2016:**
 15. **Column Chart:** Uses vertical rectangular columns to compare items across categories over time.
 16. **Bar Chart:** Uses horizontal bars; ideal when category names are long or comparisons are non-temporal.
 17. **Line Chart:** Displays continuous data points connected by straight lines; best for showing trends over time (e.g., stock prices, temperature).
 18. **Pie Chart:** A circular disk divided into proportional slices; shows the relationship of parts to a whole for a **single data series** (percentages).
 19. **Area Chart:** Highlights the magnitude of change over time by shading the area below the line.
 20. **Scatter (XY) Chart:** Plots pairs of numerical values to show correlations or scientific relationships.
 21. **Modifying and Customising Charts:**
 22. **Chart Tools Contextual Tabs:** When a chart is clicked, two tabs appear on the Ribbon: **Design** and **Format**.
 23. **Chart Elements Button (+):** Quick shortcut floating beside the chart to add/remove Data Labels, Axis Titles, and Legends.
 24. **Moving a Chart:** Click **Move Chart** in the Location group on the Design tab to move the chart to a dedicated **Chart Sheet** or another worksheet.
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Textbook Exercises — Chapter 2

A. Tick (✓) the correct option:

1. **The entire area occupied by a chart and all its components is called the:**
2. a. Plot Area **b. Chart Area [✓]** c. Gridline d. Legend
3. **Which chart type is best suited to display trends over continuous time intervals?**
4. a. Pie Chart **b. Line Chart [✓]** c. Bar Chart d. Scatter Chart
5. **The horizontal axis of a column chart is known as the:**
6. **a. Category Axis (X-Axis) [✓]** b. Value Axis c. Legend Axis d. Data Axis
7. **Which chart displays only a single data series as slices of a circle?**
8. a. Column Chart b. Bar Chart **c. Pie Chart [✓]** d. Area Chart
9. **Excel's tool for modifying a chart type is located under which tab?**
10. **a. Design Tab [✓]** b. View Tab c. Review Tab d. Home Tab

B. Fill in the blanks:

1. A **chart** is an effective visual representation of worksheet data.
2. The **Legend** helps identify different data series by assigning distinct colors or patterns.
3. The vertical axis that displays numerical values is called the **Value Axis (Y-Axis)**.
4. **Data Labels** provide the exact values directly on the plotted bars or points.

5. In a Column Chart, values are plotted along the **vertical** axis and categories along the **horizontal** axis.

C. State True (T) or False (F):

1. A Pie chart can be used to compare multiple data series simultaneously. $\rightarrow$ [F] *(Pie charts can represent only one single data series).*
2. The Plot Area contains the chart title and legends. $\rightarrow$ [F] *(The Plot Area contains only the plotted data series and gridlines).*
3. Changing data in the worksheet automatically updates the chart. $\rightarrow$ [T]
4. A Bar chart displays data using horizontal bars. $\rightarrow$ [T]
5. Gridlines cannot be removed once a chart is created. $\rightarrow$ [F] *(They can be added or removed using the Chart Elements button).*

D. Differentiate between the following:

1. **Column Chart vs. Bar Chart:**
2. **Column Chart:** Displays data using **vertical columns**. The category axis is horizontal (X-axis) and the value axis is vertical (Y-axis).
3. **Bar Chart:** Displays data using **horizontal bars**. The category axis is vertical (Y-axis) and the value axis is horizontal (X-axis).
4. **Chart Area vs. Plot Area:**
5. **Chart Area:** The complete rectangular space encompassing all chart elements (title, legends, axes, and plot area).
6. **Plot Area:** The inner region bounded by the axes where the actual graphical data (bars, lines, slices) is rendered.

E. Answer the following questions in brief:

Q1: What are the main components of a chart in Excel?

Answer: The primary components are: Chart Area, Plot Area, Chart Title, Category Axis (X-axis), Value Axis (Y-axis), Data Series, Legend, Gridlines, and Data Labels.

Q2: How can you move an existing chart to another worksheet?

Answer: Click on the chart $\rightarrow$ Go to the **Design Tab** under Chart Tools $\rightarrow$ Click **Move Chart** in the Location group $\rightarrow$ In the Move Chart dialog box, choose either **New sheet** or select an existing worksheet from the **Object in** dropdown $\rightarrow$ Click **OK**.

Chapter 3: Advanced Features of Excel 2016 (Pages 30–41)

Core Theoretical Concepts:

1. **Sorting Data:**
2. Organizing data in a specific sequential order to analyze it efficiently.
3. **Sort Orders:**
 - Ascending Order: Smallest to largest (Numbers) or A to Z (Text).
 - Descending Order: Largest to smallest (Numbers) or Z to A (Text).
4. **Custom / Multi-Level Sorting:** Sorting data by more than one column (e.g., first sort by Department in alphabetical order, and within each department, sort by Salary from largest to smallest). Located on **Data Tab** $\rightarrow$ **Sort**.
5. **Filtering Data:**
6. Filtering extracts and displays only the records that meet specified criteria, temporarily hiding unwanted rows.
7. Unlike sorting, filtering does not rearrange data; it selectively hides non-matching rows.
8. **AutoFilter:** Added by selecting headers and clicking **Data Tab** $\rightarrow$ **Filter** (Shortcut: **Ctrl + Shift + L**). Small dropdown arrows appear on each column header.
9. **Filter Types:**
 - Number Filters: *Equals, Does Not Equal, Greater Than, Less Than, Between, Top 10*.
 - Text Filters: *Equals, Contains, Begins With, Ends With*.
10. **Conditional Formatting:**
11. Allows users to automatically apply specific formatting (fill colors, font bolding, borders, data bars, icon sets) to cells that fulfill certain criteria.
12. Located on: **Home Tab** $\rightarrow$ **Styles Group** $\rightarrow$ **Conditional Formatting**.
13. Presets include:
 - Highlight Cells Rules (*Greater Than, Less Than, Between, Equal To, Text that Contains*).
 - Top/Bottom Rules (*Top 10 Items, Top 10%, Below Average*).
 - Data Bars, Color Scales, and Icon Sets.

Textbook Exercises — Chapter 3

A. Tick (✓) the correct option:

1. **Arranging data in ascending or descending order is known as:**
2. a. Filtering **b. Sorting [✓]** c. Formatting d. Validating
3. **Which feature selectively displays rows meeting specified criteria while hiding the rest?**
4. **a. Filtering [✓]** b. Sorting c. AutoSum d. Grouping

5. **The Filter button is available under which tab on the Ribbon?**
6. a. Insert Tab b. Home Tab **c. Data Tab [✓]** d. Review Tab
7. **Conditional Formatting option is found in which group on the Home Tab?**
8. a. Font Group b. Alignment Group **c. Styles Group [✓]** d. Editing Group
9. **Custom sorting allows sorting data by:**
10. a. Only one column **b. Multiple columns [✓]** c. Font color only d. None of these

B. Fill in the blanks:

1. Arranging data from A to Z is known as **ascending** order.
2. The keyboard shortcut to turn AutoFilter on or off is **Ctrl + Shift + L**.
3. **Conditional Formatting** dynamically changes the appearance of cells based on conditions you specify.
4. When a filter is applied, the dropdown arrow changes to a **filter funnel icon**.
5. Custom sorting is configured using the **Sort Dialog Box** on the Data Tab.

C. State True (T) or False (F):

1. Filtering permanently deletes rows that do not match the criteria. $\rightarrow$ **[F]** (*It only temporarily hides them*).
2. You can apply Conditional Formatting to highlight students scoring above 90 marks in green. $\rightarrow$ **[T]**
3. Data Bars in conditional formatting visually represent values directly inside the cells. $\rightarrow$ **[T]**
4. Multi-level sorting cannot sort both text and numbers simultaneously. $\rightarrow$ **[F]** (*You can sort by text in Level 1 and number in Level 2*).
5. To clear a filter, click 'Clear' in the Sort & Filter group on the Data tab. $\rightarrow$ **[T]**

D. Answer the following questions in brief:

Q1: What is the primary difference between Sorting and Filtering?

Answer:

- **Sorting** rearranges the physical order of all records in the table (e.g., alphabetical or numerical order) without hiding any data.
- **Filtering** displays only the rows that satisfy specific criteria and temporarily hides all non-matching rows.

Q2: What is Conditional Formatting? Give one real-life use case.

Answer: Conditional Formatting is an Excel tool that applies visual styling (colors, data bars, icons) to cells automatically when their values meet designated rules.

Use Case: Highlighting student test scores where scores below 35 are automatically shaded in red, and scores of 90 or above are shaded in green.

Chapter 4: Introduction to Adobe Animate 2020 (Pages 42–55)

Core Theoretical Concepts:

1. **What is Adobe Animate 2020?**
2. Adobe Animate is a professional multimedia vector animation authoring software used to create interactive 2D vector animations, banner advertisements, cartoons, animated web graphics, and interactive educational content.
3. **Workspace Components of Adobe Animate:**
4. **Menu Bar:** Top of the screen providing access to commands (File, Edit, View, Insert, Modify, Text, Commands, Control, Window, Help).
5. **Stage:** The central white rectangular canvas where graphics, drawings, animations, and text are placed. Only elements placed within the stage boundaries appear in the exported movie.
6. **Pasteboard:** The gray background area surrounding the stage. Used to store objects, work on elements off-screen, or position items that will animate into the stage later.
7. **Timeline:** The vital panel (usually at the bottom) that organizes and controls the animation's timing, frame rate, layers, and sequencing.
8. **Layers:** Transparent sheets stacked on top of one another, allowing different animated objects to be drawn and manipulated independently without interfering with each other.
9. **Frames & Keyframes:**
 - **Frame:** Individual unit of time represented by small rectangular boxes along the timeline.
 - **Keyframe:** A critical frame (marked with a black solid dot) where a new symbol appears, changes position, or initiates an action. Shortcut: F6.
 - **Blank Keyframe:** A keyframe with no content on the stage (marked with an empty circle). Shortcut: F7.
10. **Tools Panel:** Contains all creative drawing, painting, selection, and transformation tools (Selection, Free Transform, Rectangle, Oval, Brush, Eraser, Eyedropper).
11. **Properties Panel:** Contextual inspector displaying properties (size, color, stroke weight, font, coordinates) of whatever tool or object is currently selected.
12. **Library Panel:** Central storage locker containing reusable graphic symbols, buttons, movie clips, sound effects, and imported bitmaps. Shortcut: Ctrl + L.
13. **Strokes and Fills:**
14. **Stroke:** The outer border or outline of a shape. Attributes include Stroke Color, Stroke Width, and Stroke Style.
15. **Fill:** The interior body color, gradient, or pattern inside a closed shape.
16. **Eyedropper Tool (I):** Used to sample and pick colors or stroke styles directly from existing stage objects.
17. **Symbols & Animation:**
18. A **Symbol** is an original graphic, button, or movie clip stored once in the Library and reused multiple times as **Instances** on the Stage.

19. Converting an object to a Symbol: Select object $\rightarrow$ Press **F8** (or Modify $\rightarrow$ Convert to Symbol) $\rightarrow$ Name symbol and choose type (**Movie Clip**, **Button**, or **Graphic**).
 20. **Text Animation:**
 - Type text with the Text Tool (T).
 - Break text apart into individual letters using **Ctrl + B**.
 - Convert each letter into a graphic symbol (F8).
 - Distribute to layers and apply **Create Classic Tween** or **Create Motion Tween** between keyframes.
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Textbook Exercises — Chapter 4

A. Tick (✓) the correct option:

1. **Adobe Animate 2020 is primarily used for creating:**
2. a. Word documents b. Spreadsheets **c. 2D Vector Animations [✓]** d. Operating Systems
3. **The rectangular white area where graphics and animations are created is called the:**
4. **a. Stage [✓]** b. Pasteboard c. Timeline d. Library
5. **The panel that controls the timing and layers of an animation is the:**
6. a. Tools Panel **b. Timeline [✓]** c. Properties Panel d. Swatches
7. **Which tool is used to scale, rotate, or skew an object freely?**
8. a. Selection Tool **b. Free Transform Tool (Q) [✓]** c. Hand Tool d. Zoom Tool
9. **The shortcut key to convert a selected object into a Symbol in Adobe Animate is:**
10. a. F5 b. F6 **c. F8 [✓]** d. F9
11. **The tool used to pick and copy colors from one object to another is the:**
12. a. Paint Bucket **b. Eyedropper Tool [✓]** c. Brush Tool d. Pencil Tool

B. Fill in the blanks:

1. The **Stage** is the visible drawing canvas in Adobe Animate.
2. The gray area surrounding the stage is known as the **Pasteboard**.
3. A **Keyframe** is a special frame that signifies a change in the animation, marked by a black solid circle.
4. **Layers** act like transparent sheets of plastic stacked over each other.
5. In Adobe Animate, **Stroke** refers to the outline of a shape, while **Fill** refers to the interior color.
6. Reusable animation assets stored in the Library are called **Symbols**.

C. State True (T) or False (F):

1. Objects drawn on the Pasteboard will be visible in the final exported movie. $\rightarrow$ **[F]** (*Only objects placed on the Stage are visible*).
2. A keyframe is inserted using the F6 shortcut key. $\rightarrow$ **[T]**

3. Stroke refers to the interior color of a shape. $\rightarrow$ [F] (*Stroke is the outline; Fill is the interior*).
4. The Library panel stores symbols, movie clips, and audio files. $\rightarrow$ [T]
5. Breaking apart text is done using the shortcut **Ctrl + B**. $\rightarrow$ [T]

D. Answer the following questions in brief:

Q1: What is the difference between the Stage and the Pasteboard?

Answer:

- **The Stage** is the rectangular central area where the animation takes place; all objects placed here are visible in the exported output.
- **The Pasteboard** is the gray work area surrounding the Stage where assets are kept off-screen before animating into view, or where work-in-progress objects are stored without appearing in the final movie.

Q2: What is the purpose of Keyframes in the Timeline?

Answer: Keyframes indicate a major change or action in an animation—such as a change in position, color, shape, or scale of a symbol. A keyframe with content is indicated on the timeline by a solid black circle.

Q3: Define Stroke and Fill in Adobe Animate.

Answer:

- **Stroke:** The outer border, line, or perimeter outline of an object or shape.
- **Fill:** The solid color, gradient, or pattern that fills the interior closed area of an object.

E. Answer the following questions in detail:

Q1: Describe the major components of the Adobe Animate 2020 workspace.

Answer:

1. **Stage:** The primary white canvas where animation is viewed and rendered.
2. **Pasteboard:** Off-stage working area to hold temporary objects.
3. **Timeline:** Organizes sequence, frame rates, and timing with layers and frames.
4. **Tools Panel:** Contains 2D vector drawing, coloring, and transformation tools.
5. **Properties Panel:** Context-sensitive inspector that changes depending on what is selected.
6. **Library Panel:** Manages all reusable symbols, graphics, and imported media assets.

Q2: How do you animate text in Adobe Animate? Outline the step-by-step process.

Answer:

1. Select the **Text Tool (T)** and type the desired word on the Stage.
2. Press **Ctrl + B** (Break Apart) to convert the text block into individual letter graphics.
3. Select the letters, right-click, and select **Distribute to Layers**.
4. Convert each letter into a Graphic Symbol by pressing **F8**.
5. In the Timeline, click on a future frame (e.g., Frame 30) and insert Keyframes (**F6**).
6. Move or rotate the letters to their final positions on Frame 30.

7. Right-click between Frame 1 and Frame 30 and choose **Create Classic Tween** to produce smooth motion.

Test Paper - 1: Comprehensive Assessment (Chapters 1 to 4) (Pages 56–58)

Complete Section-by-Section Solutions:

Section A: Multiple Choice Questions (MCQs)

1. Which function calculates the average of numbers in a range?
a. `AVG()` b. `AVERAGE()` [✓] c. `MEAN()` d. `SUM()`
3. In Excel, an absolute reference is denoted by:
a. % b. # c. \$ [✓] d. &
5. The chart type that represents data as horizontal bars is called:
a. Column Chart b. Bar Chart [✓] c. Line Chart d. Pie Chart
7. Which feature in Excel rearranges rows based on alphabetical or numerical order?
a. Filtering b. Sorting [✓] c. AutoSum d. Chart
9. In Adobe Animate, what is the default shortcut to create a Keyframe?
a. F5 b. F6 [✓] c. F7 d. F8
11. The tool in Adobe Animate used to modify shapes and rotate objects is the:
a. Text Tool b. Paint Tool c. Free Transform Tool [✓] d. Hand Tool

Section B: Fill in the Blanks

1. The `COUNT()` function in Excel counts only the cells that contain numbers.
2. The **Legend** in a chart identifies what each color or pattern represents.
3. The shortcut key to apply or remove AutoFilter in Excel is **Ctrl + Shift + L**.
4. In Adobe Animate, the **Pasteboard** is the off-screen workspace surrounding the stage.
5. Pressing **Ctrl + B** breaks apart grouped objects or text into editable shapes.

Section C: Match the Following

Item in Column A	Match	Definition in Column B
1. AutoSum	\$\rightarrow\$ c	Automatically inserts the <code>=SUM()</code> formula
2. Line Chart	\$\rightarrow\$ d	Displays trends and variations over continuous time
3. Conditional Formatting	\$\rightarrow\$ a	Formats cells dynamically based on specified rules

Item in Column A	Match	Definition in Column B
4. Keyframe	$\rightarrow$ e	Timeline frame indicating a change in animation (F6)
5. Eyedropper Tool	$\rightarrow$ b	Samples and copies color attributes between objects

SA-1 Computer Science Quick Revision Matrix

Concept	Key Tool / Formula	Shortcut	Primary Purpose
Sum of Range	=SUM(A1:A10)	Alt + =	Instant total of contiguous numbers
Average	=AVERAGE(A1:A10)	—	Arithmetic mean calculation
Max / Min	=MAX() / =MIN()	—	Highest and lowest value identification
Absolute Cell	=\$A\$1	F4	Locks reference during formula copy
Chart Creation	Insert $\rightarrow$ Recommended Charts	F11 (Chart Sheet)	Visual graphical analysis of data
Filter Toggle	Data $\rightarrow$ Filter	Ctrl + Shift + L	Hides non-matching records
Convert to Symbol	Modify $\rightarrow$ Convert to Symbol	F8	Creates reusable Library asset
Insert Keyframe	Insert $\rightarrow$ Timeline $\rightarrow$ Keyframe	F6	Defines motion change in animation

Concept	Key Tool / Formula	Shortcut	Primary Purpose
Break Apart	Modify $\rightarrow$ Break Apart	Ctrl + B	Deconstructs text into vector paths